

Master thesis report

LONGEST INDUCED CYCLES IN GRAPHS WITH
BOUNDED DEGREE

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Related work

- Exact Methods for the Longest Induced Cycle Problem
 - Ahmad T. Anaqreh · Boglárka G.-Tóth · Tamás Vinkó
- Induced Cycles in Graphs
 - Michael A. Henning, Felix Joos, Christian Lowenstein, and Thomas Sasse

Algorithm for finding longest induced cycle in regular graphs

Input

- Number of vertices n , neighbours for each index from 0 to $n-1$

- Example:

8

1 2 3 4

0 2 3 4

0 1 5 6

0 1 5 7

0 1 6 7

2 3 6 7

2 4 5 7

3 4 5 6

Output

Detected regular degree $d = 4$

Execution time: 0 ms Longest

Induced Cycle (length 5): 7 3 0 2 6

Machine learning

Trained ML model to test whether it could approximate or replace the exact algorithm

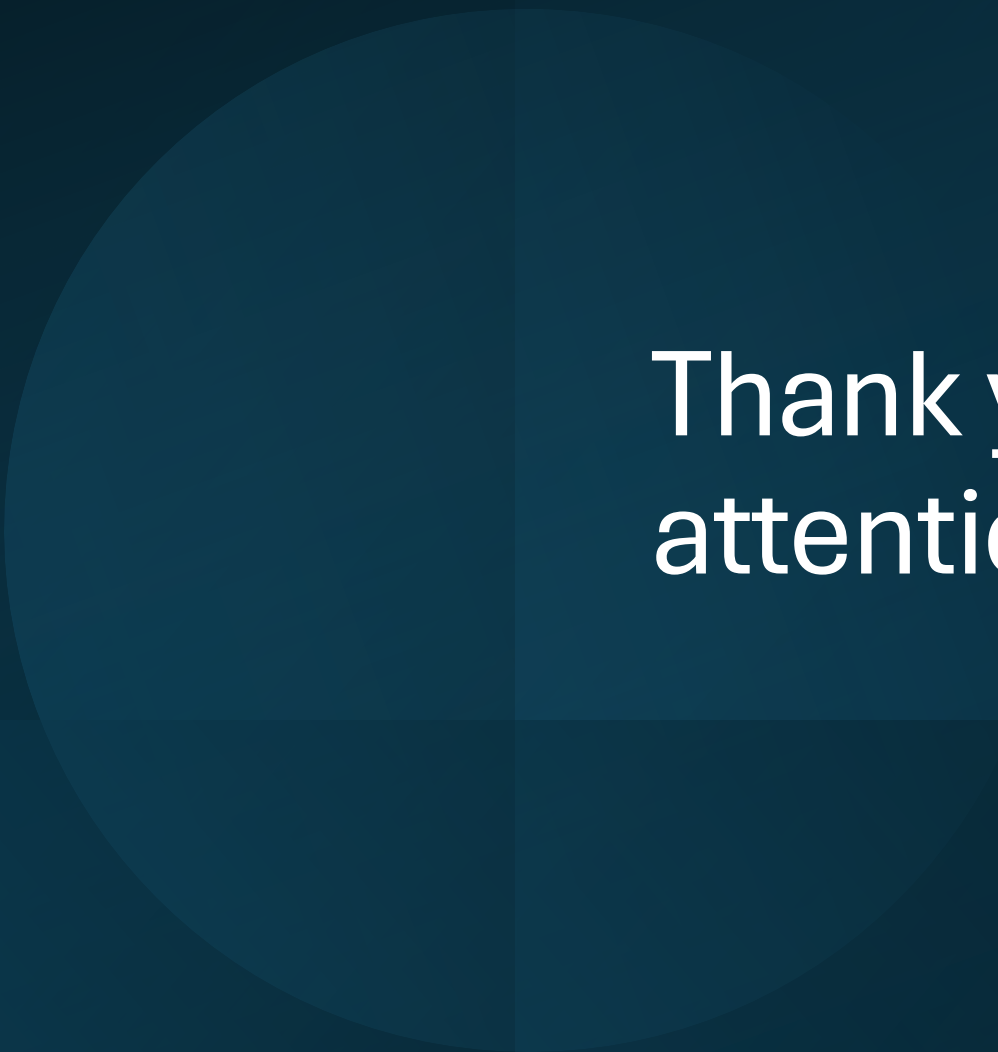
Test how much information about the problem is already contained in a few simple graph descriptors

It gives insight but remains only as a secondary tool

Theory

Proved general
lower bound
using girth

Proved general
upper bound
using the
structure



Thank you for your
attention